



## Original article

# A deep reinforcement learning cooperative air combat method with temporal feature and attention enhancement for heterogeneous flight vehicles

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## ABSTRACT

To address the challenge of joint operations involving manned aerial vehicles (MAVs), unmanned aerial vehicles (UAVs), and missiles, this paper proposes a decision-making method for air combat maneuvers among heterogeneous aircraft, termed Temporal Attention Masked Heterogeneous Agent Proximal Policy Optimization (TAM-HAPPO). The framework extends Proximal Policy Optimization (PPO) to heterogeneous multi-agent settings through a sequential update scheme. A state memory module based on gated recurrent units (GRUs) is introduced to perform temporal modeling of observation sequences and extract implicit features characterizing the dynamic battlefield. In addition, a masking mechanism and entropy regularization are incorporated into the policy network to exclude inactive agents and promote policy diversity, while a multi-head attention mechanism is integrated into the global value network to enhance critical information extraction. The algorithm is evaluated in both homogeneous and heterogeneous air combat environments. Experimental results demonstrate that TAM-HAPPO enables effective collaborative decision-making among heterogeneous aircraft, validating its robustness and applicability in complex air combat simulation scenarios.

## 1. Introduction

With the rapid advancement of information technology and UAV technology, modern air combat has undergone a qualitative change. Among them, cooperative decision-making of MAVs, UAVs, and missiles is the key to the future application of UAVs in actual air combat [1–3]. MAVs have stronger platform performance and strategic value, while UAVs have better cost performance and response speed. How to achieve coordinated decision-making between MAVs and UAVs to form complementary advantages is a key technology that urgently needs to be developed [4,5].

Cooperative air combat between MAVs and UAVs involves technologies such as mission planning, decision-making, and result evaluation. Among these, maneuvering decision-making is key to achieving victory in combat. They require aircraft to select appropriate flight control commands based on the current situation, seize advantageous positions, find opportunities to shoot down enemy aircraft, and protect themselves. To achieve this goal, the aircraft must be capable of perceiving and understanding complex air combat situations and making optimal decisions under various constraints [6,7].

From the perspective of air combat maneuver decision-making scenarios, most existing studies focus on homogeneous UAV combat, where UAVs share identical dynamic models and functional roles. According to the number of participating agents, these scenarios can be categorized into one-on-one engagements [8–11] and multi-to-one or multi-to-multi confrontations [12–16]. Due to limited payload, sensing capability, and maneuverability, a single UAV is often insufficient to accomplish complex combat missions, making cooperative multi-UAV scenarios more practically relevant. Representative studies have investigated coordinated attack formations [12], collaborative pursuit–evasion strategies [13], and large-scale swarm confrontation using mean-field approximations [14]. To address scalability, attention-based architectures that integrate local and global information have been introduced to enable effective decision-making in large-scale air combat scenarios [15,16]. Overall, current research on air combat maneuvering decision-making has achieved significant progress in homogeneous UAV swarms, while extending these methods to heterogeneous combat scenarios remains an open challenge.

Some researchers have also explored UAV maneuvering decision-making in heterogeneous environments, although such studies remain

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relatively limited. Liu et al. [17] proposed an adaptive communication and dynamic consensus mechanism for heterogeneous UAV swarms, effectively alleviating issues such as target-selection chaos in adversarial environments. Zhu et al. [18] investigated a heterogeneous UAV-missile environment and proposed a PPO-based method with incentive curriculum learning to enable advanced maneuvering and missile evasion. Shahid et al. [19] developed an autonomous cooperative allocation approach based on the Extended Contract Network Protocol (ECNP) to support multi-directional attacks by heterogeneous UAVs. Zhang et al. [20] proposed a hierarchical reinforcement learning framework to enhance the autonomy of loyal wingmen, enabling coordinated UAV-missile operations. In practical battlefield scenarios, cooperative operations among MAVs, UAVs, and missiles are common, highlighting the necessity of further research on heterogeneous air combat maneuvering decision-making. Specifically, in heterogeneous systems involving MAVs, UAVs, and missiles, the agents exhibit significantly different flight dynamics, sensing capabilities, and tactical roles, resulting in heterogeneous state-action characteristics and asymmetric decision objectives. These factors substantially increase the complexity of cooperative decision-making.

From the perspective of air combat maneuvering decision-making methodologies, existing approaches can be broadly classified into three categories: game theory-based methods, optimization theory-based methods, and artificial intelligence-based methods.

Representative game theory-based methods include differential games, influence diagrams, and matrix games. Li et al. [21] proposed a constrained strategic game approach for multi-UAV decision-making based on situational awareness and time-sensitive information. Guo et al. [22] employed influence diagrams and Bayesian inference to derive UAV maneuvering strategies, though the method suffers from high computational complexity. Li et al. [23] introduced a dimensionality reduction-based matrix game method to accelerate strategy generation, but its scalability remains limited as the number of UAVs increases. Optimization theory-based methods formulate maneuvering decision-making as optimization problems and solve them using bio-inspired algorithms or classical techniques such as dynamic programming. Chai et al. [24] proposed a distributed model predictive control framework combined with an attention-enhanced ant colony optimization algorithm for cooperative UAV decision-making. Crumpacker et al. [25] developed an adaptive dynamic programming approach using neural network approximation. However, optimization-based methods generally suffer from limited real-time performance in large-scale and dynamic air combat scenarios. Artificial intelligence-based methods primarily include expert systems and reinforcement learning. Expert systems [26,27] rely heavily on handcrafted rules and prior knowledge, leading to high development and maintenance costs. In contrast, reinforcement learning has been increasingly applied to air combat decision-making due to its strong adaptability. Zhuang et al. [28] proposed a hybrid-action reinforcement learning method for autonomous UAV maneuvering. Xu et al. [29] introduced a value decomposition attention-based reinforcement learning framework to address cooperative multi-UAV decision-making. Wu et al. [30] proposed a context-aware feature fusion method to enhance adversarial intention recognition. Pang et al. [31] developed a hierarchical MAPPO-based framework for cooperative air combat, while Wang et al. [32] proposed an evolutionary multi-agent reinforcement learning method to mitigate policy cycling. Overall, existing methods still face significant challenges in cooperative decision-making for heterogeneous air combat systems, motivating further research in this area.

In this work, we investigate the cooperative air combat decision-making problem in heterogeneous manned-unmanned-missile systems, and propose a unified maneuvering strategy framework based on the TAM-HAPPO algorithm. The main contributions of this paper are summarized as follows:

1. We formulate a unified deep reinforcement learning-based cooperative air combat decision-making problem involving MAVs, UAVs, and missiles. In this heterogeneous environment, MAVs and UAVs possess different performance constraints, autonomy levels, and tactical objectives. Unlike most existing studies that focus on homogeneous UAV confrontation, the proposed formulation explicitly considers asymmetric roles and mixed autonomy, which makes the formulated problem more representative of heterogeneous air combat decision-making scenarios.
2. To address the non-stationarity and asymmetric policy coupling inherent in heterogeneous air combat, an air combat maneuvering decision-making method for heterogeneous aircraft is proposed based on the TAM-HAPPO algorithm. The PPO algorithm is extended to heterogeneous environments by incorporating the multi-agent advantage decomposition theorem and a sequential update scheme, enabling stable joint training and coordinated policy learning among agents with heterogeneous action spaces and tactical roles.
3. Considering the strong temporal dependency and partial observability of air combat environments, a state memory module based on a GRU architecture is designed to perform time-series modeling of observation data and generate implicit features that capture the dynamic evolution of battlefield states. The experience replay buffer is organized in temporal order to enhance time consistency and long-term situational awareness in decision-making.
4. To improve robustness and scalability in complex heterogeneous scenarios, several task-driven mechanisms are further introduced, including an agent masking strategy to exclude inactive entities, entropy regularization to prevent premature convergence, and a multi-head attention mechanism integrated into the centralized value network. These components jointly enhance the efficiency, adaptability, and robustness of cooperative decision-making in heterogeneous air combat.

The subsequent organization of this paper is as follows: [Section 2](#) describes the research problem and establishes the heterogeneous aircraft air combat environment, including aircraft, missile, and air combat situational assessment model. [Section 3](#) presents the deep reinforcement learning-based maneuvering decision-making method for heterogeneous aircraft, covering the overall framework, the design of state and action spaces, algorithm enhancements, and reward function design. [Section 4](#) details the simulation experiments, providing results and analysis for two different environments, and validates the effectiveness of the algorithm improvements through ablation studies. [Section 5](#) concludes the paper.

## 2. Problem description and modeling

### 2.1. Problem description

This paper designs a heterogeneous air combat environment that includes MAVs, UAVs, and missiles. Our side forms a combat formation composed of one MAV and several UAVs, carrying missiles that follow fixed guidance laws to engage enemy UAVs. The objective is to achieve air combat victory by maximizing enemy aircraft kills while ensuring the safety of the MAV. Unlike traditional homogeneous UAV confrontation environments, the MAV and UAVs in this scenario have different performance parameters and tactical roles, introducing greater challenges to cooperative decision-making. Specifically, MAVs and UAVs exhibit heterogeneous flight performance constraints, sensing capabilities, and survivability requirements, which lead to different operational behaviors and decision preferences. In addition, the agents assume asymmetric tactical roles in cooperative air combat. The MAV mainly focuses on mission coordination and survivability, while UAVs perform cooperative engagement and support tasks. Such heterogeneous dynamics and

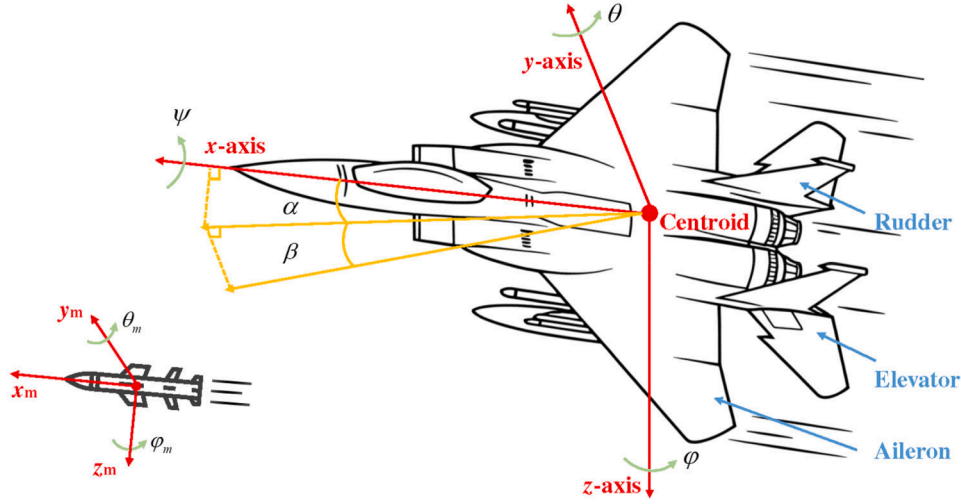


Fig. 1. Aircraft and missile models.

role asymmetry significantly increase the complexity of designing unified cooperative decision policies compared with conventional homogeneous UAV scenarios. It should be noted that the current environment focuses on capturing the key characteristics of heterogeneous cooperative air combat for algorithmic evaluation. More detailed operational factors such as communication constraints, sensing uncertainties, and control delays are left for future investigation.

## 2.2. Aircraft dynamics

In order to accurately simulate the motion behavior of aircraft and missiles within an air combat environment, a dynamic model for the aircraft and a kinematic model for the missile are developed in this study, as shown in Fig. 1. First, establish two coordinate systems: the earth-fixed coordinate system  $O_e X_e Y_e Z_e$  and the body-fixed coordinate system  $O_b X_b Y_b Z_b$ . The earth-fixed coordinate system is used to describe the position and orientation of the aircraft's center of gravity relative to the ground. The origin  $O_e$  can be selected at any point on the ground. The body-fixed coordinate system is fixed on the aircraft, and its origin  $O_b$  is located at the aircraft's center of gravity.

The speed vector of aircraft can be represented as  $V = (u, v, w)$ , where  $u, v, w$  represents the projection of the aircraft's speed vector on the three body-fixed coordinate axes, and the speed scalar of aircraft is  $|V| = \sqrt{u^2 + v^2 + w^2}$ . The angle of attack is denoted as  $\alpha$ , representing the angle between the projection of the velocity vector on the body-fixed coordinate plane  $O_b X_b Z_b$  and vector  $O_b X_b$ , and can be obtained from Eq. (1).

$$\alpha = \arctan \frac{w}{u}, \quad -\pi \leq \alpha \leq \pi \quad (1)$$

The sideslip angle is denoted as  $\beta$ , representing the angle between the velocity vector and the body-fixed coordinate plane  $O_b X_b Z_b$ , and can be obtained from Eq. (2).

$$\beta = \arcsin \frac{v}{|V|}, \quad -\pi \leq \beta \leq \pi \quad (2)$$

The projections of the velocity vector onto the three body-fixed coordinate axes can be obtained from Eq. (3).

$$\begin{cases} u = |V| \cos \alpha \cos \beta \\ v = |V| \sin \beta \\ w = |V| \sin \alpha \cos \beta \end{cases} \quad (3)$$

The six degrees of freedom of a fixed-wing aircraft include three translational degrees of freedom (corresponding to the aircraft's longitudinal, lateral, and vertical movements) and three rotational degrees of

freedom (corresponding to the aircraft's pitch angle  $\theta$ , yaw angle  $\varphi$ , and roll angle  $\psi$ ). The acceleration equation of the aircraft in the aircraft coordinate system is given by Eq. (4).

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = \begin{bmatrix} u \\ v \\ w \end{bmatrix} \times \begin{bmatrix} p \\ q \\ r \end{bmatrix} + \frac{1}{m} \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} + C_g^b \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} \quad (4)$$

where,  $p, q, r$  are the angular velocities around the  $x, y, z$  axes of the body-fixed coordinate system,  $F_x, F_y, F_z$  represent the external forces acting on the three coordinate axes,  $m$  is the mass of the aircraft,  $g$  is the gravitational acceleration with a value of  $9.8m/s^2$ , and  $C_g^b$  is the transformation matrix from the earth-fixed coordinate system to the body-fixed coordinate system, given by Eq. (5).

$$C_g^b = \begin{pmatrix} \cos \varphi \cos \theta & \sin \varphi \cos \theta & -\sin \theta \\ \cos \varphi \sin \theta \sin \psi - \sin \varphi \cos \psi & \sin \varphi \sin \theta \sin \psi + \cos \varphi \cos \psi & \cos \theta \sin \psi \\ \cos \varphi \sin \theta \cos \psi + \sin \varphi \sin \psi & \sin \varphi \sin \theta \cos \psi - \cos \varphi \sin \psi & \cos \theta \cos \psi \end{pmatrix} \quad (5)$$

The angular acceleration equation in the body-fixed coordinate system is given by Eq. (6).

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = I^{-1} \left( - \begin{bmatrix} p \\ q \\ r \end{bmatrix} \times I \begin{bmatrix} p \\ q \\ r \end{bmatrix} + \begin{bmatrix} M_x \\ M_y \\ M_z \end{bmatrix} \right) \quad (6)$$

where,  $I$  is the inertia matrix of the aircraft, reflecting the distribution characteristics of mass in the body-fixed coordinate system,  $M_x, M_y, M_z$  are the external torques acting on the three coordinate axes.

## 2.3. Missile dynamics and guidance law

The missile motion is modeled in the inertial coordinate system using a point-mass assumption. The kinematic equations of the missile position are given by Eq. (7).

$$\begin{cases} \dot{x}_m = V_m \cos \theta_m \cos \varphi_m \\ \dot{y}_m = V_m \cos \theta_m \sin \varphi_m \\ \dot{z}_m = V_m \sin \theta_m \end{cases} \quad (7)$$

where  $(x_m, y_m, z_m)$  denote the missile position,  $V_m$  is the velocity magnitude, and  $(\theta_m, \varphi_m)$  are the pitch and yaw angles, respectively.

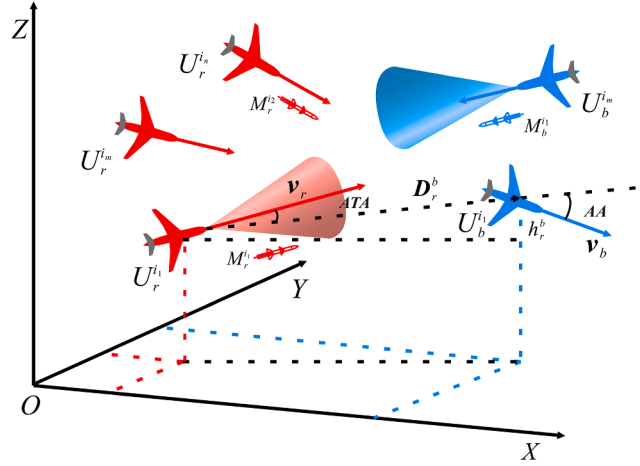


Fig. 2. Geometric posture of air combat coordination between MAVs, UAVs, and missiles.

The missile velocity and attitude dynamics are governed by Eq. (8).

$$\begin{cases} \dot{V}_m = g(n_x - \sin \theta_m), \\ \dot{\varphi}_m = \frac{g}{V_m} n_y \cos \theta_m, \\ \dot{\theta}_m = \frac{g}{V_m} (n_z - \cos \theta_m), \end{cases} \quad (8)$$

where  $g$  is the gravitational acceleration,  $n_x$  denotes the longitudinal overload, and  $n_y, n_z$  represent the lateral overloads in the yaw and pitch directions, respectively. During the active phase,  $n_x$  accounts for thrust and aerodynamic drag, while in the passive phase, the missile is affected only by gravity and drag.

A proportional navigation (PN) guidance law is adopted to generate lateral overload commands. The guidance commands are expressed as Eq. (9).

$$\begin{cases} n_y = \frac{K_y V_m}{g} \cos \theta_m \dot{\theta}_m, \\ n_z = \frac{K_z V_m}{g} \dot{\varphi}_m + \cos \theta_m, \end{cases} \quad (9)$$

where  $K_y$  and  $K_z$  are the navigation gains, both set to 3 in this work.  $\dot{\theta}_m$  and  $\dot{\varphi}_m$  denote the time derivatives of the line-of-sight (LOS) angles in the yaw and pitch directions, which are computed from the relative position and velocity between the missile and the target.

#### 2.4. Air combat situation assessment

In an air combat environment, the tactical situation between two opposing sides can generally be classified into three categories: our side holds the advantage, the enemy holds the advantage, or both sides are in a state of equilibrium. The situation assessment in this study is constructed based on five key indicators: antenna train angle (ATA), aspect angle (AA), relative altitude ( $h_r^b$ ), relative distance ( $D_r^b$ ), and relative velocity ( $v_r^b$ ). These parameters jointly characterize the geometric relationship and kinematic state between opposing aircraft, providing an intuitive basis for engagement evaluation.

$n$  red-team aircraft engage with  $m$  blue-team UAVs, as illustrated in Fig. 2. Taking  $U_r^i$  and  $U_b^j$  as an example,  $v_r$  and  $v_b$  denote the velocity vectors of the red and blue aircraft, respectively.  $D_r^b$ ,  $h_r^b$ , and  $v_r^b$  represent the relative position vector, altitude difference, and velocity difference between the two aircraft. ATA is defined as the angle between the red aircraft's velocity vector and the line-of-sight (LOS) vector, while AA denotes the angle between the blue aircraft's velocity vector and the extension of the LOS vector.

Analyze the geometric relationships of the air combat situation in Fig. 2. The relevant parameters are calculated using Eq. (10).

$$\begin{cases} v_r^b = v_r - v_b \\ h_r^b = z_r - z_b \\ \begin{cases} \text{ATA} = \arccos \frac{D_r^b \cdot v_r}{\|D_r^b\| \|v_r\|} \\ \text{AA} = \arccos \frac{D_r^b \cdot v_b}{\|D_r^b\| \|v_b\|} \end{cases} \end{cases} \quad (10)$$

where,  $D_r^b = (x_b - x_r, y_b - y_r, z_b - z_r)$ ,  $(x_r, y_r, z_r)$  and  $(x_b, y_b, z_b)$  are the coordinates of the red and blue teams in the three-dimensional coordinate system.

Based on the above parameters, the air combat situation assessment function can be divided into four aspects: angle, distance, altitude, and speed, as shown in Eq. (11).

$$\begin{cases} e_a = 1 - \frac{\text{ATA} + \text{AA}}{2\pi} \\ e_d = \mathbb{I}(d \leq d_{\max}) \\ e_h = \frac{h_r^b}{h_{\max}} \\ e_v = \frac{\|v_r^b\|}{v_{\max}} \end{cases} \quad (11)$$

where  $d_{\max}$  denotes the maximum effective engagement range of the missile,  $\mathbb{I}(\cdot)$  is the indicator function, and  $h_{\max}$  and  $v_{\max}$  are predefined normalization constants.

By assigning different weights to each component, the overall situational assessment score is obtained as:

$$E_{ij} = \omega_a e_a^{(ij)} + \omega_v e_v^{(ij)} + \omega_h e_h^{(ij)} + \omega_d e_d^{(ij)} \quad (12)$$

where  $E_{ij}$  represents the situational advantage of aircraft  $i$  relative to aircraft  $j$ . In this study, the weighting coefficients are empirically initialized as  $\omega_a = 0.35$ ,  $\omega_d = 0.25$ ,  $\omega_h = 0.20$ , and  $\omega_v = 0.20$ .

It should be emphasized that the situation assessment function is introduced as an auxiliary mechanism for target selection and engagement prioritization. The weighting coefficients are fixed throughout training and do not participate in policy optimization or reward computation. Consequently, the initialization of these coefficients does not directly bias the learning process, while enabling efficient reduction of multi-target decision complexity.

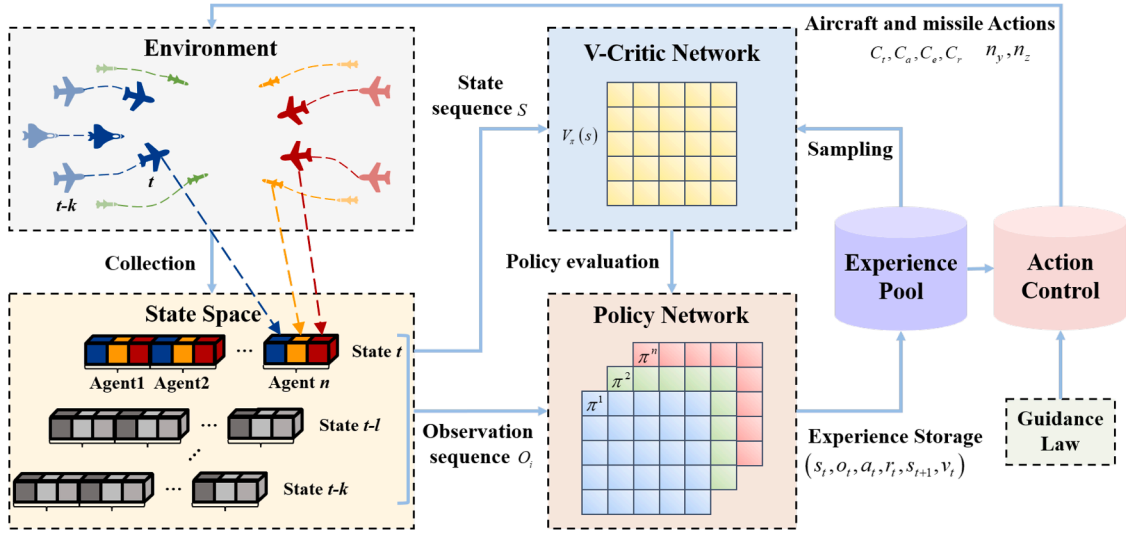


Fig. 3. Overall architecture of the cooperative combat method for heterogeneous aircraft.

### 3. Heterogeneous aircraft cooperative air combat approach

#### 3.1. Overall framework

The overall framework of the cooperative air combat maneuver decision-making method for heterogeneous aircraft based on the TAM-HAPPO algorithm is illustrated in Fig. 3. The framework consists of two main components: the environment layer and the algorithm layer. The environment layer is responsible for collecting relevant situational data and encapsulating it into feature vectors, which are then input into the state memory module to further extract temporal hidden features. The algorithm layer primarily comprises the policy networks, the value function network, and the experience replay buffer. Both MAVs and UAVs are modeled as autonomous agents within this framework and participate equally in the decision-making process. Each aircraft is equipped with an independent policy network, which takes its own observation space as input and outputs the corresponding maneuvering commands. These independent policy networks are collaboratively optimized in the heterogeneous environment through a sequential update scheme. All aircraft share a common value function network, which takes the global state of the environment and the joint actions of all agents as input and outputs the estimated state value. This shared value guides the training of the policy networks and ensures coordinated decision-making among heterogeneous agents.

#### 3.2. State space and action space

In a UAV cooperative air combat environment, it is difficult to observe the global information by the UAV itself alone, which means that the state of the environment is not fully observable. To address this situation, a partially observable Markov decision process is introduced to model the UAV co-operation problem. The entire Markov decision process can be represented by a tuple  $G = (N, S, O^i, A^i, P, R^i, \gamma)$ . Where,  $N = \{1, \dots, n\}$  denotes the finite set of UAVs,  $S$  is the environmentally state space,  $O^i$  is the observation space for each UAV, and the joint action space is denoted by  $A = A^1 \times \dots \times A^n (i \in N)$ , which is a collocation of the action space  $A^i$  of all  $n$  UAVs within the red-team UAV cluster. The state transfer matrix  $P: S \times A \rightarrow P(S)$  describes the state transfer probability of a multi-intelligent system, each UAV possesses an associated reward function  $R^i$  that provides immediate guidance information, and  $\gamma \in [0, 1)$  is the discount factor.

In order to obtain comprehensive information about the positioning of both sides on the battlefield, our aircrafts are equipped with an

airborne GPS, a gyroscope and other equipment. This enables them to acquire information about their own position, speed and attitude. The heterogeneous combat formation formed when the MAV joins has enhanced its ability to perceive battlefield information and can detect enemy UAVs, determining their position, speed and other details. Considering that aircrafts must cooperate in combat while evading incoming enemy missiles, the observation space of each aircraft is composed of its own state and relative state information, which can be expressed in Eq. (13).

$$O^i = (x_i, y_i, z_i, v_i, \theta_i, \varphi_i, \psi_i, \Delta v_{ij}, \Delta h_{ij}, d_{ij}, ATA_{ij}, AA_{ij}) \quad (13)$$

where,  $(x_i, y_i, z_i)$  represent the position coordinates of aircraft  $i$  in the ground coordinate system,  $v_i$  denotes the speed of aircraft  $i$ , and  $(\theta_i, \varphi_i, \psi_i)$  correspond to the pitch angle, yaw angle, and roll angle of aircraft  $i$ , respectively. These seven dimensions constitute the self-state of the aircraft. The relative state includes  $\Delta v_{ij}$  and  $\Delta h_{ij}$ , which represent the relative speed and relative altitude between aircraft  $i$  and another entity  $j$ . Additionally,  $d$ ,  $ATA_{ij}$  and  $AA_{ij}$  denote the distance, antenna train angle, and aspect angle between the two entities. The entity  $j$  can refer to other friendly aircraft, enemy UAVs, or incoming missiles locked onto aircraft  $i$ .

The heterogeneous aircraft combat system proposed in this paper operates within a fully cooperative reinforcement learning environment. Therefore, the global state  $S$  can be represented as the aggregation of the local observations of all aircraft, which can be expressed in Eq. (14).

$$S = (O^1, O^2, \dots, O^n) \quad (14)$$

As all aircraft are fixed-wing, the maneuvering control can be achieved through four variables: throttle control  $C_t$ , aileron  $C_a$ , elevator  $C_e$  and rudder  $C_r$ . The action space of each UAV is  $A_i = [C_t, C_a, C_e, C_r]$ , where  $C_t$  has a value range of  $[0.4, 0.9]$  and  $C_a, C_e, C_r$  has a value range of  $[-1, 1]$ . To balance simulation accuracy with computational complexity, the continuous action space is represented in a multi-discrete form, with 40 discrete levels per action dimension.

#### 3.3. HAPPO algorithm framework

To adapt to heterogeneous air combat environments involving multiple aircraft and missiles, and to enhance the collaborative decision-making capabilities of intelligent agents in high-dynamic, multi-type, and large-scale scenarios, this paper proposes a collaborative air combat decision-making method based on the TAM-HAPPO algorithm. This method is applicable to multi-agent systems, including both homogeneous and heterogeneous systems, and possesses high scalability. It can



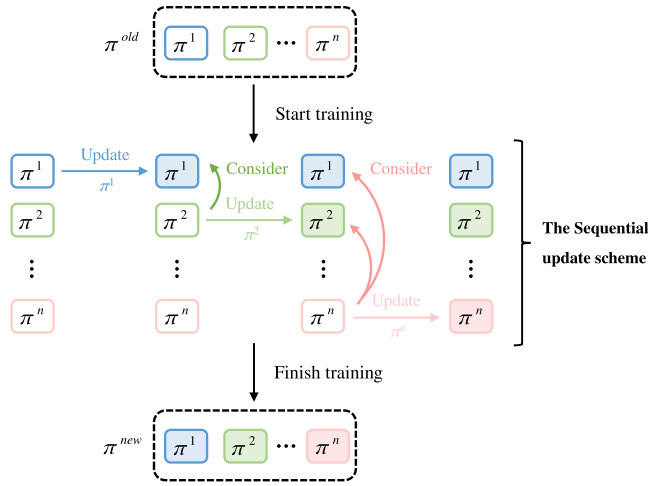


Fig. 4. Multi-agent strategy sequential updating scheme.

simultaneously handle collaborative combat tasks involving multiple aircraft of different models, significantly enhancing the overall intelligence level and combat effectiveness of the air combat system.

The HAPPO algorithm is a version of the PPO algorithm extended to multi-agent systems [33]. Unlike the previous MAPPO algorithm, the HAPPO algorithm is applicable to both homogeneous agents and heterogeneous agent environments. Its core lies in proposing the multi-agent advantage decomposition lemma and strategy sequence update scheme. The multi-agent advantage function as shown in Eq. (15).

$$A_{\pi}^{i1:m}(s, \mathbf{a}^{i1:k}, \mathbf{a}^{i1:m}) \triangleq Q_{\pi}^{j1:k,i1:m}(s, \mathbf{a}^{i1:k}, \mathbf{a}^{i1:m}) - Q_{\pi}^{j1:k}(s, \mathbf{a}^{i1:k}) \quad (15)$$

where,  $A_{\pi}^{i1:m}(s, \mathbf{a}^{i1:k}, \mathbf{a}^{i1:m})$  represents the relative advantage of UAV  $i1:m$  taking action  $\mathbf{a}^{i1:m}$  after UAV  $j1:k$  takes action  $\mathbf{a}^{i1:k}$  in state  $s$ ,  $Q_{\pi}^{j1:k,i1:m}(s, \mathbf{a}^{i1:k}, \mathbf{a}^{i1:m})$  represents the value of UAV  $i1:m$  taking action  $\mathbf{a}^{i1:m}$  after UAV  $j1:k$  takes action  $\mathbf{a}^{i1:k}$ , and  $Q_{\pi}^{j1:k}(s, \mathbf{a}^{i1:k})$  represents the average value of UAV  $j1:k$  taking action  $\mathbf{a}^{i1:k}$ . The multi-agent advantage decomposition is given by Eq. (16).

$$A_{\pi}^{i1:m}(s, \mathbf{a}^{i1:m}) = \sum_{j=1}^m A_{\pi}^{i,j}(s, \mathbf{a}^{i1:j-1}, \mathbf{a}^{i,j}) \quad (16)$$

Eq. (16) proves that in a heterogeneous multi-agent environment, the strategy of each agent can be updated sequentially. If the advantage function of each agent can be maximized, then a joint optimal strategy can be found, and each update can take into account the information of the updated strategy. The multi-agent advantage decomposition and sequence update process is shown in Fig. 4.

The main structure of the HAPPO algorithm consists of  $n$  policy networks and a global value function network. Each UAV has an independent policy network, with the input being the observation space of each UAV and the output being the action distribution of each UAV. The input of the global value function network is the joint observation space, and the output is the value under the joint policy. The loss function of the policy network is shown in Eq. (17).

$$\begin{cases} \mathcal{L}_{\text{Actor}} = \frac{1}{BT} \sum_{b=1}^B \sum_{t=0}^T \min \left( r_t^{i_m} A_t^{i1:m}(s_t, \mathbf{a}_t), \text{clip}(r_t^{i_m}, 1 \pm \epsilon) A_t^{i1:m}(s_t, \mathbf{a}_t) \right) \\ r_t^{i_m} = \frac{\pi_{\theta^{i_m}}^m(\mathbf{a}_t^{i_m} | \mathbf{o}_t^{i_m})}{\pi_{\theta_k^m}^m(\mathbf{a}_t^{i_m} | \mathbf{o}_t^{i_m})} \end{cases} \quad (17)$$

where,  $B$  is the batch size,  $T$  is the step size of each trajectory,  $i_m$  represents the  $m$ th UAV,  $m \in [1, n]$ ,  $r_t^{i_m}$  is the ratio of the new and old policies, and  $A_t^{i1:m}(s_t, \mathbf{a}_t)$  is the advantage function of the HAPPO algorithm,

which is calculated by Eq. (18).

$$A^{i1:m+1}(s, \mathbf{a}) = \frac{\pi_{\theta_{k+1}^m}^m(\mathbf{a}^{i_m} | \mathbf{o}^{i_m})}{\pi_{\theta_k^m}^m(\mathbf{a}^{i_m} | \mathbf{o}^{i_m})} A^{i1:m}(s, \mathbf{a}) \quad (18)$$

Eq. (18) indicates that after each UAV completes its strategy update, the advantage function is recalculated until  $m = n$ , where  $n$  is the total number of UAVs.

The loss function of the global value function network is shown in Eq. (19).

$$\mathcal{L}_{\text{Value}} = \frac{1}{BT} \sum_{b=1}^B \sum_{t=0}^T (V_{\phi}(s_t) - \hat{R}_t)^2 \quad (19)$$

where,  $V_{\phi}(s_t)$  is the value function network, and  $\hat{R}_t$  is the reward estimate used in the advantage function.

### 3.4. Network structure and optimization

#### 3.4.1. State memory module design

In actual air combat, environmental conditions have obvious temporal correlations, and UAVs that rely solely on current observations often fail to form stable and reliable strategies. To enhance the UAV's perception and understanding of battlefield dynamics, this paper designs a state memory module (SM) that introduces a gated recurrent neural network (GRU) to model observation sequences and store long-term dependencies, thereby improving the temporal modeling and generalization capabilities of strategies. Its structure is shown in Fig. 5.

GRU is a variant of recurrent neural networks that controls information flow through update gates and reset gates. Its core structure is given by Eq. (20).

$$\begin{cases} z_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \\ r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \\ \tilde{h}_t = \tanh(W_h \cdot [r_t * h_{t-1}, x_t]) \\ h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \end{cases} \quad (20)$$

To ensure temporal consistency during training, we designed a time-aligned experience replay mechanism. Each training batch contains observation information from multiple consecutive time steps, which is input into the state memory module in order. The GRU then outputs the corresponding hidden state, which is sent to the subsequent model together with the state sequence information from the next time step, until all historical states have been processed. Before being input into the policy network, the generated hidden state features are first transformed into the specified state dimension via a linear layer to avoid excessive dimension after feature concatenation, which would consume computational resources. The policy network with the state memory module can be represented by Eq. (21)

$$A_t = MLP(\varpi_t | \theta) \quad (21)$$

where,  $MLP(\cdot)$  represents a multilayer perceptron (MLP) with two layers, whose activation functions are two fully connected layers (FC),  $\varpi_t$  represents the concatenated information input to the policy network after passing through the SM module, and  $\theta$  represents the weights that the policy network needs to learn.

The global value network with the state memory module can be represented by Eq. (22).

$$V_t = MLP((\varpi_t^V \oplus A_t) | \phi) \quad (22)$$

where, the symbol  $\oplus$  represents a cascade operation,  $\varpi_t^V$  represents the concatenated information input to the value network after passing through the SM module, and  $\phi$  represents the weights that the value network needs to learn.

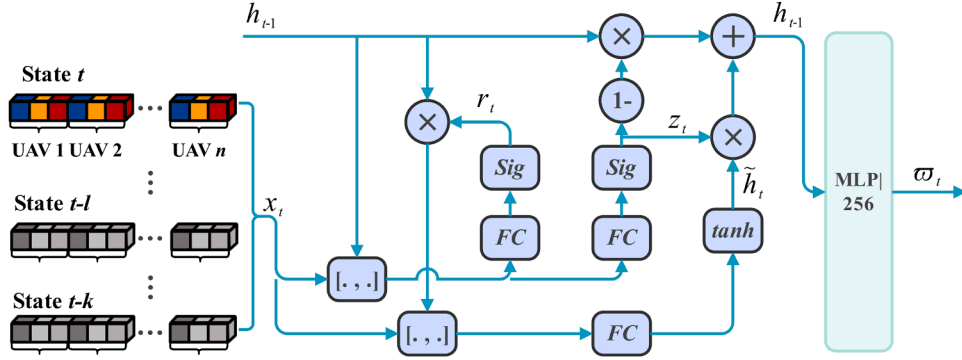


Fig. 5. Structure of the state memory module.

### 3.4.2. Policy network based on masking and entropy regularization

In a volatile air combat environment, some UAVs may be shot down by the enemy. Traditional strategy networks cannot handle the inconsistency of survival status and continue to pass the strategy output of dead agents to the joint strategy optimizer. This may cause interference, thereby impairing the learning and execution of other surviving agents.

To address the aforementioned challenges, this paper introduces a masking mechanism into the policy network, which consists of two components. The first component masks out inactive agents. During both training and decision-making, it dynamically identifies UAVs that are no longer active and excludes their contributions from policy gradient computation and joint policy evaluation. The second component masks out unavailable actions. In heterogeneous multi-UAV and multi-missile environments, agents may have different action spaces depending on their types. This action masking ensures consistency in policy modeling while allowing for flexible configuration of heterogeneous action spaces. We introduce a mask scalar  $k_t^i \in \{0, 1\}$  for each UAV  $i^m$ , which indicates whether the UAV is still alive at that time step. After adding the mask mechanism, the loss function of the policy network can be expressed by Eq. (23).

$$\mathcal{L}_{\text{happo}} = \frac{1}{BT} \sum_{b=1}^B \sum_{t=0}^T k_t^i \cdot \min \left( r_t^i A_t^{1:m}(s_t, a_t), \text{clip}(r_t^i, 1 \pm \epsilon) A_t^{1:m}(s_t, a_t) \right) \quad (23)$$

In addition, to prevent the algorithm from converging to local optima and to enhance exploratory capability during the early stages of training, this paper incorporates an entropy regularization term  $\mathcal{H}[\pi(a | o)]$  into the loss function of the policy network. This encourages the policy to produce a more uniform distribution over actions, promoting exploration of diverse behaviors. As a result, UAVs are more likely to discover novel strategies and improve overall robustness, while maintaining a certain level of policy uncertainty to reduce predictability by adversaries. The loss function for the entropy regularization term is given by Eq. (24).

$$\mathcal{L}_{\text{entropy}} = \frac{1}{BT} \sum_{b=1}^B \sum_{t=0}^T k_t^i \cdot \rho \cdot \left( - \sum_a \pi_\theta(a_t^i | o_t^i) \log \pi_\theta(a_t^i | o_t^i) \right) \quad (24)$$

where,  $\rho$  is the coefficient controlling the strength of the entropy regularization term.

The final loss function of the policy network is the sum of the above two losses, as shown in Eq. (25). And the structure of the policy network is shown in Fig. 6(a).

$$\mathcal{L}_{\text{Actor}} = \mathcal{L}_{\text{happo}} + \mathcal{L}_{\text{entropy}} \quad (25)$$

### 3.4.3. Value network based on multi-head attention

In existing reinforcement learning algorithms, the value function network typically employs a MLP to directly model the global state and

joint actions, while neglecting the potential interaction between states and actions. This limitation hinders the learning of cooperative strategies in multi-agent systems. To address this issue, this paper introduces an embedding layer, a multi-head attention mechanism, and Layer Normalization (Layer Norm) into the value function network, thereby restructuring the network and enhancing its ability to capture the coupled relationships between states and actions. The improved architecture of the value network is illustrated in Fig. 6(b).

The input to the network is the concatenation of the global states and the joint action  $a$ . This combined input is first processed through an embedding layer, which maps the input vector into a unified feature space, facilitating subsequent computations in the attention mechanism. The calculation of the embedding layer is shown in Eq. (26).

$$h_t = W_e x_t + b_e \quad (26)$$

where,  $x_t$  is the input vector,  $W_e$  is the embedding weight matrix, and  $b_e$  is the bias term.

The feature vector  $h_t$  is used as the new input and is projected through linear layers using distinct projection matrices  $W^Q, W^K, W^V$  to generate  $k$  sets of matrices, where  $k$  represents the number of attention heads. Each set contains three matrices: the Query matrix, the Key matrix, and the Value matrix. The computation process is given by Eq. (27).

$$Q = h_t W^Q, \quad K = h_t W^K, \quad V = h_t W^V \quad (27)$$

Subsequently, the  $k$  sets of matrices are passed into the attention layer for attention computation, which is the core of the multi-head attention mechanism. Each attention head produces an output vector  $H_i (i = 1, 2, \dots, k)$ . The computation process as shown in Eq. (28).

$$H_i = \text{Attention}(Q_i, K_i, V_i) = \text{softmax} \left( \frac{Q_i K_i^T}{\sqrt{d_k}} \right) V_i \quad (28)$$

The outputs from all attention heads are concatenated through the Concat layer, and then passed through a final linear transformation to obtain the output vector  $y$ , which has the same dimensionality as the input feature  $h_t$ . The operation process is given by Eq. (29).

$$y = \text{MultiHead}(Q, K, V) = \text{Concat}(H_1, H_2, \dots, H_m) W^0 \quad (29)$$

After completing the attention mechanism computation, the output vector is passed through two MLP layers with LayerNorm applied after each layer. Finally, the processed features are passed through a linear layer to produce the output of the value network.

### 3.5. Reward function

In the reinforcement learning framework, the intelligent agent interacts with the environment in real time and receives rewards or penalties to evaluate the quality of its actions. By learning from these experiences, the agent aims to make optimal decisions that maximize cumulative

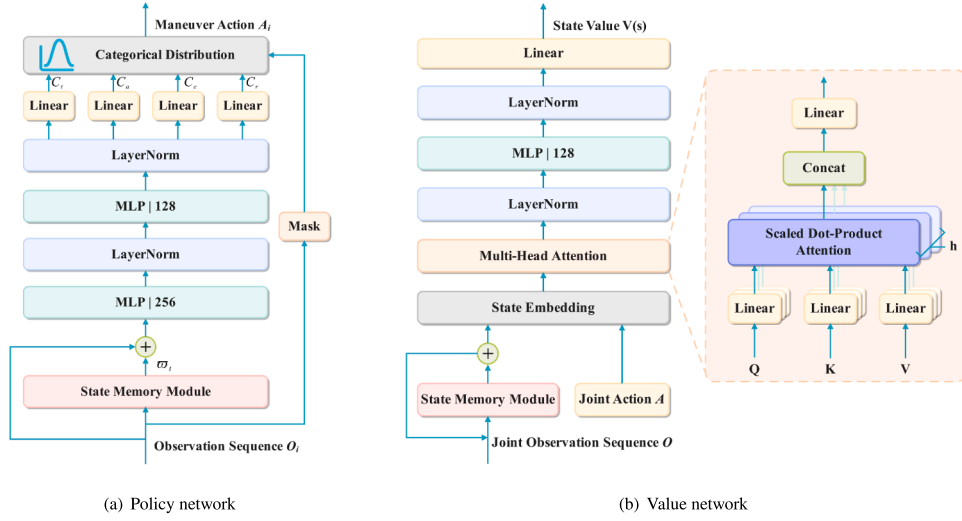


Fig. 6. The structures of the policy and value networks.

Table 1

Detailed reward function design for MAV and UAV agents.

| Agent | Reward Module   | Mathematical Formulation                                                                                                                                                                                                                                                                                                                                                                           | Design Objective                                          |
|-------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| MAV   | Safety Reward   | $R_{\text{safety}} = 0.5R_{\text{dist}} + 0.3R_{\text{threat}} + 0.2R_{\text{aspect}}$ $R_{\text{dist}} = \begin{cases} -\left(1 - \frac{d}{d_{\text{danger}}}\right), & d < d_{\text{danger}} \\ -0.5\left(1 - \frac{d - d_{\text{danger}}}{d_{\text{safe}} - d_{\text{danger}}}\right), & d_{\text{danger}} \leq d < d_{\text{safe}} \\ 0.2, & d \geq d_{\text{safe}} \end{cases}$               | Encourage survivability and threat avoidance              |
|       |                 | $R_{\text{threat}} = -\mathbb{I}(\text{under missile attack})$                                                                                                                                                                                                                                                                                                                                     |                                                           |
|       |                 | $R_{\text{aspect}} = -\sum \mathbb{I}\left(TA < \frac{\pi}{4}\right) \left(1 - \frac{TA}{\pi/4}\right)$                                                                                                                                                                                                                                                                                            |                                                           |
| UAV   | Support Reward  | $R_{\text{support}} = 0.6R_{\text{pos}} + 0.4R_{\text{aware}}$ $R_{\text{pos}} = \begin{cases} \frac{d_b}{d_{\text{opt}}} - 1, & d_b < d_{\text{opt}} \\ 1 - \frac{d_b - d_{\text{opt}}}{d_{\text{max}} - d_{\text{opt}}}, & d_{\text{opt}} \leq d_b < d_{\text{max}} \\ -0.5, & d_b \geq d_{\text{max}} \end{cases}$ $R_{\text{aware}} = \sum_{AO < \pi/2} 0.3 \left(1 - \frac{AO}{\pi/2}\right)$ | Maintain supportive positioning and battlefield awareness |
|       | Event Reward    | $R_{\text{event}} = -\mathbb{I}(\text{death}) \cdot C_d + \sum \min(C_k, C_{\text{max}})$                                                                                                                                                                                                                                                                                                          | Enforce survival priority and bounded team contribution   |
|       | Height Reward   | $R_H = P_V + P_H$                                                                                                                                                                                                                                                                                                                                                                                  | Maintain safe flight altitude                             |
|       | Speed Reward    | $R_V = \begin{cases} 1, & V_b < 0.5V_r \\ 2 - 2\frac{V_b}{V_r}, & 0.5V_r \leq V_b \leq 1.5V_r \\ -1, & V_b \geq 1.5V_r \end{cases}$                                                                                                                                                                                                                                                                | Encourage advantageous velocity control                   |
|       | Angle Reward    | $R_A = 1 - \frac{ATA + AA}{\pi}$                                                                                                                                                                                                                                                                                                                                                                   | Promote favorable engagement geometry                     |
|       | Distance Reward | $R_D = \begin{cases} 1, & R \leq 5 \\ e^{-0.921(R-5)}, & 5 < R < 10 \\ -1, & R \geq 10 \end{cases}$ $R_{DM} = R_{AM} + R_{SM}$                                                                                                                                                                                                                                                                     | Guide effective engagement distance                       |
|       | Dodge Reward    | $R_{AM} = -\cos \lambda$ $R_{SM} = \frac{V_{M(t-1)} - V_{M(t)}}{V_{\text{norm}}}$                                                                                                                                                                                                                                                                                                                  | Improve missile evasion capability                        |
|       | Event Reward    | $R_E = \begin{cases} 200, & \text{kill enemy} \\ -200, & \text{shot down/crash} \\ -100, & \text{out of combat zone} \end{cases}$                                                                                                                                                                                                                                                                  | Handle sparse combat outcomes                             |

rewards. The reward functions for MAV and UAVs are summarized in Table 1.

For MAVs, the reward design emphasizes survivability, situational awareness, and team-level coordination. The safety reward integrates distance-based threat evaluation, missile warning penalties, and unfavorable aspect-angle penalties to discourage exposure to high-risk combat geometries. The support reward encourages MAVs to maintain ad-

vantageous positions relative to the evolving battlefield center while preserving sufficient awareness of enemy aircraft, thereby enabling effective command, coordination, and support functions. The altitude reward constrains the flight envelope by penalizing unsafe low-altitude maneuvers and incentivizing operation near an optimal altitude band. In addition, the event reward imposes a strong penalty on manned aircraft loss and introduces a capped team contribution bonus, ensuring



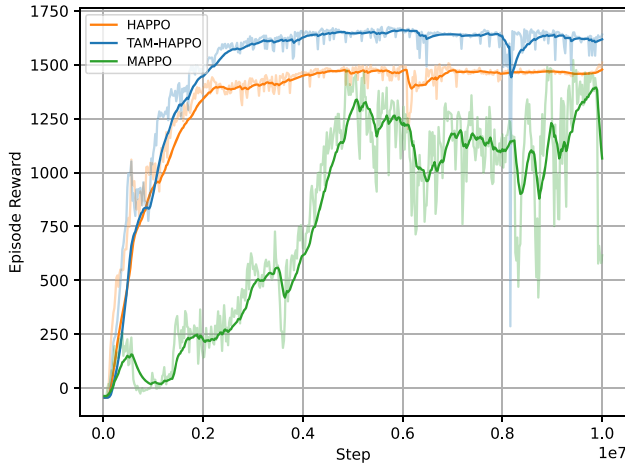


Fig. 7. Reward curve in a 2v2 homogeneous environment.

**Table 2**  
Aircraft performance parameters.

| Parameter                   | Value      |
|-----------------------------|------------|
| Maximum Flight Speed (m/s)  | 400        |
| Minimum Flight Altitude (m) | 750        |
| Maximum Overload (g)        | 9          |
| Throttle $C_t$              | [0.4, 0.9] |
| Aileron $C_a$               | [-1, 1]    |
| Elevator $C_e$              | [-1, 1]    |
| Rudder $C_r$                | [-1, 1]    |

that survival remains the highest priority and preventing reward inflation due to excessive team-based gains.

For UAVs, the reward function focuses on aggressive maneuvering, engagement effectiveness, and local survivability. Height and speed rewards regulate basic flight safety and kinematic advantage, while angle and distance rewards guide UAVs toward favorable missile engagement geometries and effective firing ranges. The dodge reward explicitly accounts for missile evasion by encouraging velocity attenuation and unfavorable missile approach angles. Event-based rewards are introduced to handle sparse but critical combat outcomes, such as successful kills, vehicle losses, and boundary violations.

## 4. Experimental results and analysis

### 4.1. Experimental environment setup

The hardware environment for the experiments in this paper used Intel(R) Core(TM) i5-12400 CPU @2.50 GHz, 16 GB RAM and GeForce RTX 3090 graphics card. The experiment was developed using VSCode, with a Python 3.8 language environment and the open source Pytorch 2.1.0 deep learning framework.

The experimental architecture can be divided into a physical simulation layer and an algorithm training layer. The physical simulation layer is based on JSBSIM [34], which is a general-purpose object-oriented flight dynamics model (FDM) written in C++. It can accurately simulate the motion of aircraft, missiles, and other objects in various environments. The algorithm training layer is implemented in Python. A heterogeneous intelligent agent deep reinforcement learning framework is designed to enable aircraft to make intelligent decisions in a complex environment involving multiple aircraft and missiles.

### 4.2. Model parameter setup

The experiment was conducted in a many-to-many air combat environment, where both sides engaged in tactical maneuvers within a desig-

**Table 3**  
Missile performance parameters.

| Parameter            | Value  |
|----------------------|--------|
| Maximum Overload (g) | 30     |
| Attack Range (m)     | 14,000 |
| Attack Interval (s)  | 25     |
| Mass (kg)            | 84     |
| Length (m)           | 2.87   |
| Diameter (m)         | 0.127  |

**Table 4**  
TAM-HAPPO algorithm parameters.

| Parameter                  | Value      |
|----------------------------|------------|
| Total training steps       | 10,000,000 |
| Training steps per episode | 1000       |
| GRU units                  | 128        |
| MLP units                  | [256, 128] |
| Policy learning rate       | 0.0005     |
| Value learning rate        | 0.0005     |
| PPO clip                   | 0.2        |
| Gradient clip              | 10         |
| Entropy coefficient        | 0.01       |
| GAE factor                 | 0.95       |
| Discount factor            | 0.99       |
| Huber loss coefficient     | 10         |

nated airspace. The red-side aircraft adopted the TAM-HAPPO algorithm proposed in this paper for maneuver decision-making, whereas the blue-side aircraft employed a greedy algorithm for their control strategy. Specifically, the blue-side controller was implemented as a rule-based finite-state machine, in which a set of basic maneuver actions was pre-defined in advance. At each decision step, the blue-side aircraft evaluated all candidate maneuvers according to reward function and greedily selected the action that maximized the instantaneous reward, following the methodology described in [8]. This rule-based strategy provides a stable and interpretable adversarial policy, which is commonly used in air combat decision-making studies as a benchmark for evaluating different learning algorithms under consistent opponent behavior.

In the physical simulation layer, JSBSim was configured to run at a frequency of 60 Hz, while maneuver decisions were made at 5 Hz. This indicates that the decision-making algorithm updated the selected actions every 0.2 s, with each action maintained for 12 simulation steps. The detailed parameters of the aircraft and missiles are provided in Tables 2 and 3, respectively. Such a greedy baseline serves as a stable and interpretable opponent, which facilitates a clear evaluation of the learning performance and adaptability of the proposed TAM-HAPPO algorithm.

In the algorithm training layer, different weight coefficients were designed to balance the importance of each internal reward. The weight coefficients for height, speed, angle, distance, and avoidance were set to 10, 10, 15, 10, and 30, respectively. During the algorithm training process, the reward, policy gradient, loss, and other relevant data of the red-team UAV are recorded every five iterations. Additionally, a test is conducted every 25 iterations to evaluate performance, during which the test reward of the red-team UAV is recorded and the corresponding model is saved. The main parameters of the TAM-HAPPO algorithm are given in Table 4. The pseudo-code of the TAM-HAPPO algorithm is shown in Algorithm 1.

### 4.3. Simulation results and analysis

This paper uses Python to implement the training and testing of algorithms. During testing, relevant aircraft and missile data is saved and tacview software is used to visualize air combat trajectories.

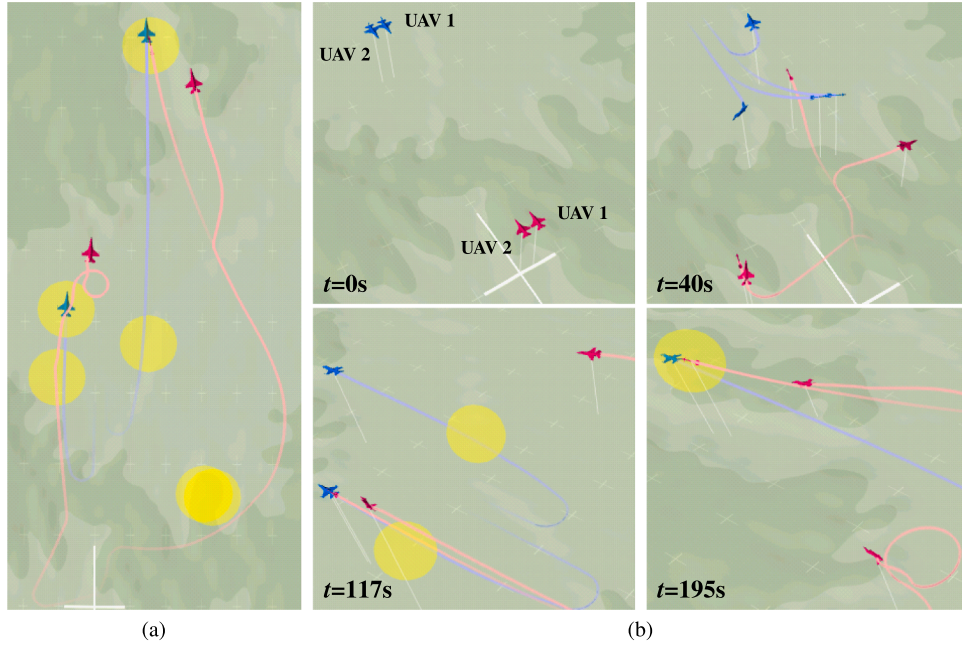


Fig. 8. Air combat trajectory in a 2v2 homogeneous environment.

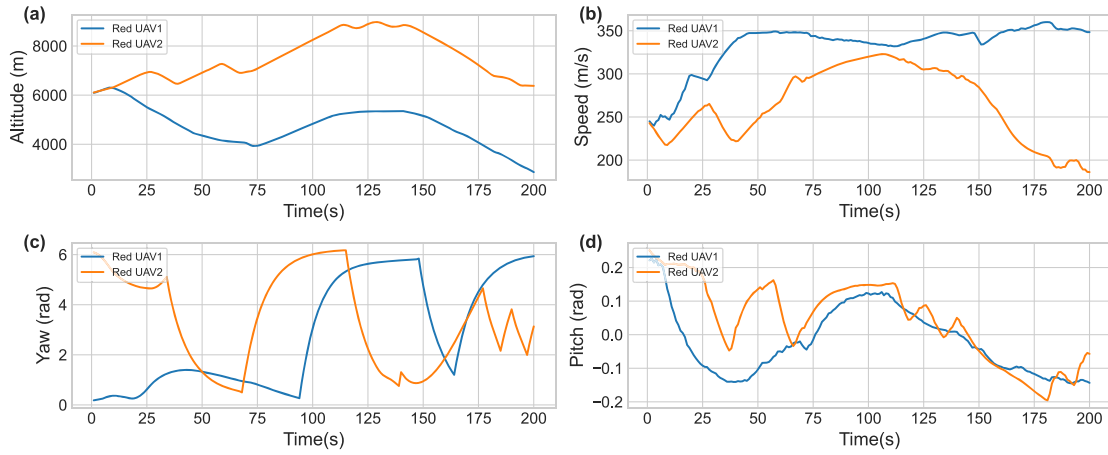


Fig. 9. Aircraft attitude change curve in 2v2 homogeneous environment.

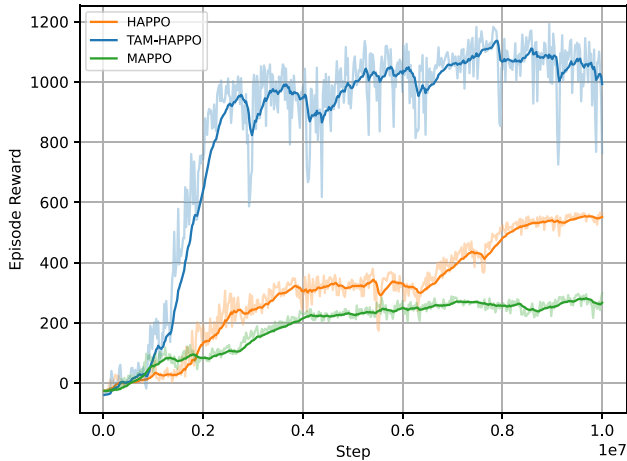


Fig. 10. Reward curve in a 3v2 heterogeneous environment.

#### 4.3.1. Homogeneous 2v2 environment

In a homogeneous 2v2 air combat environment, the red team consists of two UAVs of the same model, and the blue team has the same number and model of UAVs. At the start of the battle, the UAVs on both sides have the same altitude and speed. Each UAV is carrying two missiles and is flying towards the other team's UAVs on opposite sides of the battlefield. The initial parameters are shown in Table 5.

Fig. 7 shows the training reward curves for the TAM-HAPPO, HAPPO [33], and MAPPO [35] algorithms in a 2v2 homogeneous environment. As shown in Fig. 7, in the 2v2 homogeneous environment, the TAM-HAPPO algorithm achieves the highest reward return after training stabilizes, while HAPPO and MAPPO yield comparable performance. However, in terms of convergence speed, both TAM-HAPPO and HAPPO converge significantly faster than MAPPO. This improvement is attributed to the introduction of a sequential update mechanism, which ensures continuous and effective optimization throughout the training process.

Fig. 8 illustrates the air combat trajectory curves of the red and blue teams in this environment. At the onset of the engagement, two red-team UAVs and two blue-team UAVs fly toward each other, each carry-

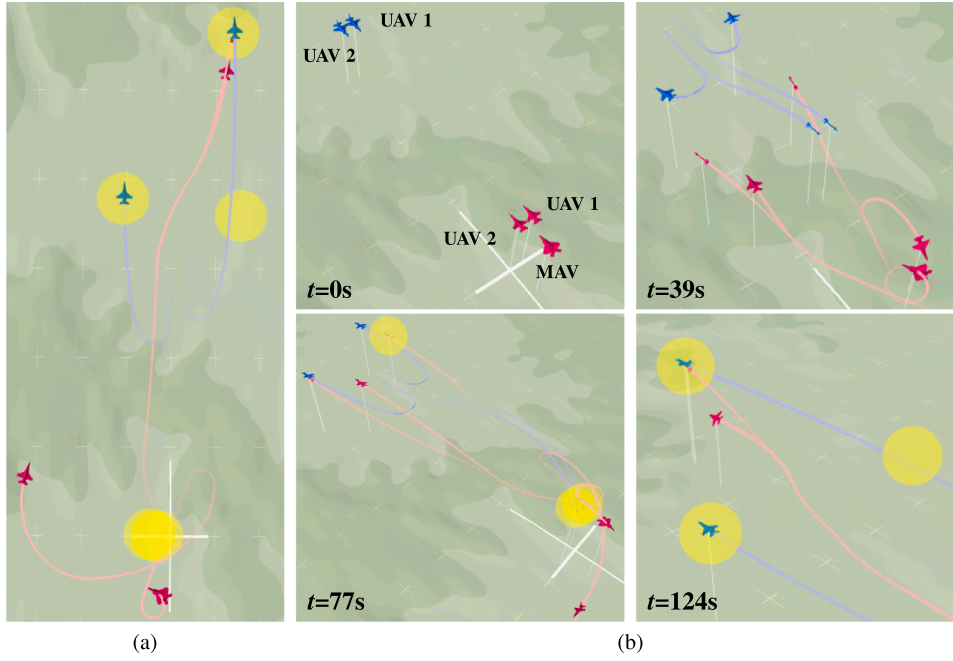


Fig. 11. Air combat trajectory in a 3v2 heterogeneous environment.

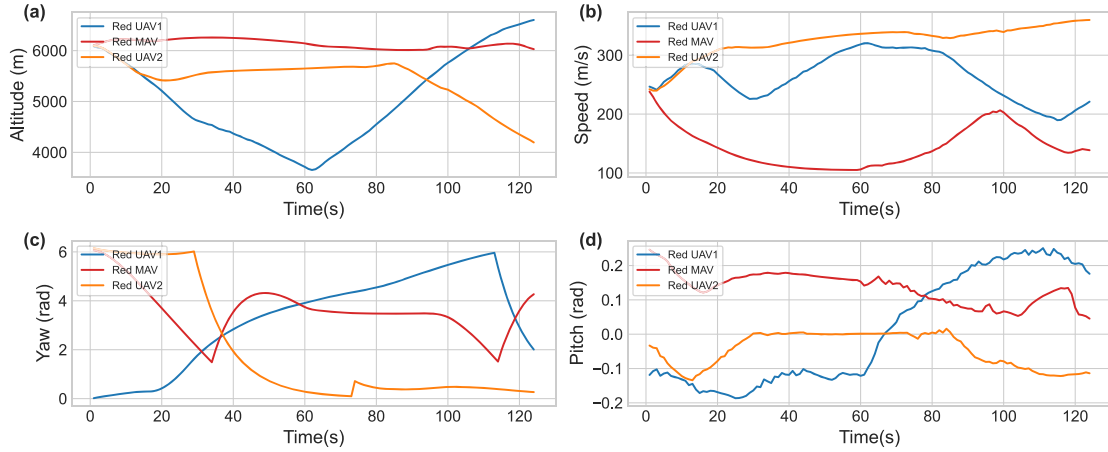


Fig. 12. Aircraft attitude change curve in 3v2 heterogeneous environment.

ing two missiles of the same type. The yellow circles indicate the missile blast radius. At  $t = 40s$ , red-team UAV 1 advances, locks onto blue-team UAV 2, launches a missile, and simultaneously draws enemy fire. It then performs a sharp right turn to evade incoming threats. Meanwhile, red-team UAV 2 maneuvers to the left flank of blue-team UAV 2, locks onto the target, and launches a missile. In response, both blue-team UAVs attempt evasive maneuvers; however, the red-team UAVs maintain a positional advantage and successfully shoot down the opponents at  $t = 117s$  and  $t = 195s$ , respectively. The engagement concludes with a decisive victory for the red team.

The air combat trajectories in Fig. 8 demonstrate that the two red-team UAVs have learned a baiting tactic, in which UAV 1 lures the enemy into firing, allowing UAV 2 to flank from the rear and side to secure victory. To gain a clearer understanding of the UAVs' attitude changes during the engagement, Fig. 9 presents the variation curves of altitude, velocity, yaw angle, and pitch angle for the two red-team UAVs in the 2v2 homogeneous environment.

#### 4.3.2. Heterogeneous 3v2 environment

To evaluate the training performance of the proposed algorithm in a heterogeneous environment, a 3v2 heterogeneous air combat scenario is designed. In this setup, the red team consists of a heterogeneous formation composed of one MAV and two UAVs, while the blue team comprises two identical UAVs. The MAV does not carry missiles and is responsible for providing battlefield information and mission guidance to the UAVs while ensuring its own safety. Each UAV starts with the same initial altitude and velocity and carries two missiles of the same type. The opposing teams fly toward each other from opposite sides of the battlefield. The initial parameters are detailed in Table 6.

Fig. 10 presents the training reward curves of the TAM-HAPPO, HAPPO, and MAPPO algorithms in the 3v2 heterogeneous environment. As shown in Fig. 10, the TAM-HAPPO algorithm achieves the highest reward after training stabilizes, significantly outperforming the other two algorithms. The MAPPO algorithm performs poorly in the

**Algorithm 1: TAM-HAPPO.**

**Input:** Number of agents  $N$ ; maximum training episodes  $E$ ; maximum episode length  $T$ ;  
 policy network  $\pi_\theta$  with parameters  $\theta$ ; value network  $V_\phi$  with parameters  $\phi$ ;  
 state memory module SM (two-layer GRU); entropy coefficient  $\rho$ ;  
 clipping parameter  $\epsilon$ ; learning rates  $\alpha_\theta, \alpha_\phi$ .  
**Output:** Optimized actor parameters  $\theta$  and critic parameters  $\phi$ .

- 1: Initialize  $\theta, \phi$ , and state memory module SM
- 2: **for**  $e \leftarrow 1$  **to**  $E$  **do**
- 3:   Reset environment and state memory SM
- 4:   Obtain initial observation  $o_1^i$  for each agent  $i$
- 5:   **for**  $t \leftarrow 1$  **to**  $T$  **do**
- 6:     **for** each agent  $i = 1, \dots, N$  **do**
- 7:        $\varpi_t^i = \text{SM}(o_t^i)$
- 8:       Concatenate  $\varpi_t^i$  with  $o_t^i$  to form input  $\delta_t^i$
- 9:       Sample action  $a_t^i \sim \pi_\theta(\cdot | \delta_t^i)$
- 10:      Apply action masking
- 11:    **end**
- 12:    Execute joint action  $\mathbf{a}_t$  and observe reward  $r_t$ , next observation  $o_{t+1}^i$
- 13:    Store transition in on-policy buffer
- 14:   **end**
- 15:   Compute advantage estimates  $\hat{A}_t^i$  using GAE
- 16:   **for** each gradient step **do**
- 17:     Compute clipped TAM-HAPPO loss:  
 $\mathcal{L}_{\text{Actor}} = \mathcal{L}_{\text{happo}} + \mathcal{L}_{\text{entropy}}$
- 18:     Update policy parameters:  $\theta \leftarrow \theta + \alpha_\theta \nabla_\theta \mathcal{L}_{\text{Actor}}$
- 19:     Update value network with multi-head attention:  
 $\phi \leftarrow \phi - \alpha_\phi \nabla_\phi \|V_\phi(\delta_t^i) - R_t^i\|^2$
- 20:   **end**
- 21: **end**

**Table 5**

Initial states under the 2v2 environment.

| Aircraft   | Initial Coordinates                  | Altitude (m) | Speed (m/s) | Yaw Angle ( $^\circ$ ) |
|------------|--------------------------------------|--------------|-------------|------------------------|
| Red UAV 1  | (120 $^\circ$ E, 60 $^\circ$ N)      | 6000         | 250         | 0                      |
| Red UAV 2  | (120.02 $^\circ$ E, 60 $^\circ$ N)   | 6000         | 250         | 0                      |
| Blue UAV 1 | (120 $^\circ$ E, 60.2 $^\circ$ N)    | 6000         | 250         | 180                    |
| Blue UAV 2 | (120.02 $^\circ$ E, 60.2 $^\circ$ N) | 6000         | 250         | 180                    |

**Table 6**

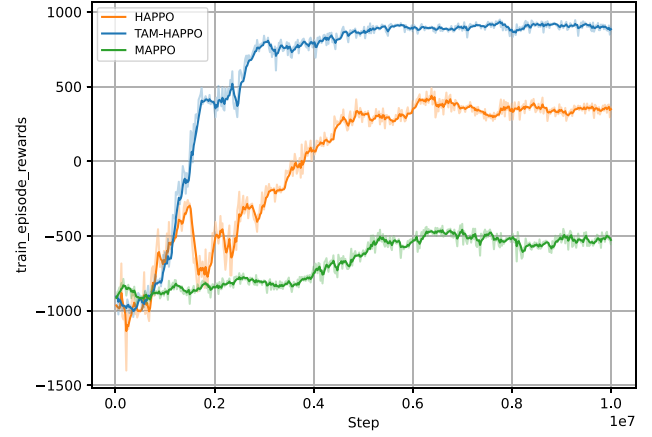
Initial states under the 3v2 environment.

| Aircraft   | Initial Coordinates                   | Altitude (m) | Speed (m/s) | Yaw Angle ( $^\circ$ ) |
|------------|---------------------------------------|--------------|-------------|------------------------|
| Red UAV 1  | (120 $^\circ$ E, 60 $^\circ$ N)       | 6000         | 250         | 0                      |
| Red UAV 2  | (120.04 $^\circ$ E, 60 $^\circ$ N)    | 6000         | 250         | 0                      |
| Red MAV 1  | (120.02 $^\circ$ E, 59.98 $^\circ$ N) | 6000         | 250         | 0                      |
| Blue UAV 1 | (120 $^\circ$ E, 60.2 $^\circ$ N)     | 6000         | 250         | 180                    |
| Blue UAV 2 | (120.04 $^\circ$ E, 60.2 $^\circ$ N)  | 6000         | 250         | 180                    |

**Table 7**

Initial states under the 5v4 environment.

| Aircraft   | Initial Coordinates                   | Altitude (m) | Speed (m/s) | Yaw Angle ( $^\circ$ ) |
|------------|---------------------------------------|--------------|-------------|------------------------|
| Red MAV 1  | (120 $^\circ$ E, 60 $^\circ$ N)       | 6700         | 250         | 0                      |
| Red UAV 1  | (119.98 $^\circ$ E, 60.02 $^\circ$ N) | 6000         | 250         | 0                      |
| Red UAV 2  | (120.02 $^\circ$ E, 60.02 $^\circ$ N) | 6000         | 250         | 0                      |
| Red UAV 3  | (119.99 $^\circ$ E, 59.98 $^\circ$ N) | 6000         | 250         | 0                      |
| Red UAV 4  | (120.01 $^\circ$ E, 59.98 $^\circ$ N) | 6000         | 250         | 0                      |
| Blue UAV 1 | (119.98 $^\circ$ E, 60.15 $^\circ$ N) | 6000         | 250         | 180                    |
| Blue UAV 2 | (120.02 $^\circ$ E, 60.15 $^\circ$ N) | 6000         | 250         | 180                    |
| Blue UAV 3 | (119.99 $^\circ$ E, 60.19 $^\circ$ N) | 6000         | 250         | 180                    |
| Blue UAV 4 | (120.01 $^\circ$ E, 60.19 $^\circ$ N) | 6000         | 250         | 180                    |

**Fig. 13.** Reward curve in a 5v4 heterogeneous environment.

heterogeneous setting due to its simultaneous update of all agents' action selections, which overlooks the differences among heterogeneous agents. In terms of convergence speed, TAM-HAPPO also demonstrates a clear advantage, converging to a reward of approximately 1000 within two million training steps.

Fig. 11 illustrates the air combat trajectories of the red and blue teams in the heterogeneous environment. At the start of the engagement, the red team forms an inverted triangle formation consisting of one MAV positioned at the rear and two UAVs at the front. The MAV is unarmed, while each UAV carries two missiles. The blue team consists of two identical UAVs, each also equipped with two missiles. As both sides fly toward each other, at  $t = 39$ s, the red team's MAV circles in the rear airspace to ensure its own safety while providing battlefield information. Guided by this information, the two red UAVs advance, lock onto enemy targets, and launch missiles. Simultaneously, the blue UAVs lock onto red UAV 2 and, after firing their missiles, attempt to retreat from the battlefield. At  $t = 77$ s, the red MAV successfully evades the incoming missile through maneuvering, while red UAV 1 shoots down blue UAV 1. Under the command of the manned aircraft, it continues to pursue blue UAV 2 and successfully destroys it at  $t = 124$ s. The engagement concludes with a victory for the red team.

The air combat trajectories in Fig. 11 indicate that the red team's heterogeneous formation successfully achieved coordinated operations: the manned aircraft provided battlefield information, while the two UAVs executed a pursuit tactic while evading incoming missiles. To gain a clearer understanding of the attitude changes of both the manned and unmanned aircraft during the engagement, Fig. 12 presents the variation curves of altitude, velocity, yaw angle, and pitch angle for the red team's manned aircraft and two UAVs in the 3v2 heterogeneous environment.

#### 4.3.3. Heterogeneous 5v4 environment

To further evaluate the scalability and robustness of the proposed algorithm in larger heterogeneous formations, a 5v4 heterogeneous air combat scenario is designed. In this setup, the red team consists of a heterogeneous formation composed of one MAV and four UAVs, while the blue team comprises four identical UAVs.

Similar to the 3v2 scenario, the MAV does not carry missiles and primarily serves as a command and support unit, providing battlefield information and mission guidance to the UAVs while prioritizing its own survivability. Each UAV is initialized with the same altitude and velocity and is equipped with two missiles of the same type.

The two opposing teams are initially deployed on opposite sides of the battlefield and fly toward each other. All other environmental settings and simulation parameters remain consistent with those in the 3v2



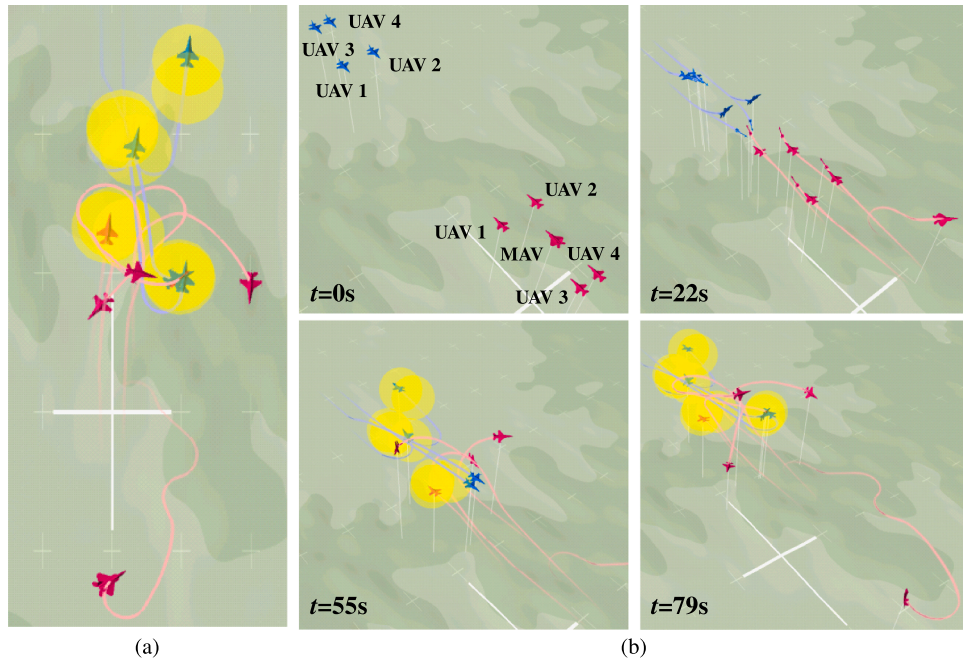


Fig. 14. Air combat trajectory in a 5v4 heterogeneous environment.

Table 8

Initial-state perturbation settings for generalization experiments (5v4 environment).

| Perturbation Level | Altitude (m) | Longitude (deg) | Latitude (deg) | Heading (°) | Velocity (m/s) |
|--------------------|--------------|-----------------|----------------|-------------|----------------|
| Low                | $\pm 500$    | $\pm 0.005$     | $\pm 0.005$    | $\pm 5$     | $\pm 20$       |
| Medium             | $\pm 1000$   | $\pm 0.01$      | $\pm 0.01$     | $\pm 10$    | $\pm 35$       |
| Large              | $\pm 1500$   | $\pm 0.015$     | $\pm 0.015$    | $\pm 15$    | $\pm 50$       |

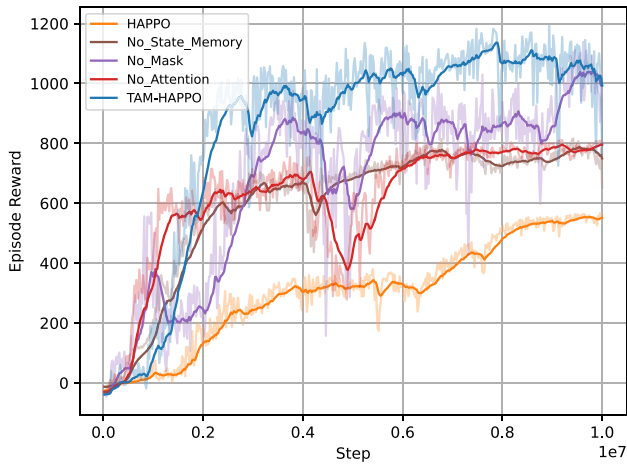


Fig. 15. Reward curve of ablation experiment.

scenario, ensuring a fair comparison across different scales. The detailed initial parameters are summarized in Table 7.

Fig. 13 shows the training reward curves of TAM-HAPPO, HAPPO, and MAPPO in the 5v4 heterogeneous air combat environment. Due to the increased number of agents and more complex coordination requirements, this scenario is more challenging than the 3v2 case. In all experiments, the adversarial aircraft follow the same rule-based greedy policy described in Section 4.2, which provides a consistent evaluation benchmark for different learning algorithms. As illustrated in the figure, TAM-HAPPO achieves the highest final reward and demonstrates

faster and more stable convergence compared with the other two algorithms. HAPPO converges more slowly with noticeable fluctuations, while MAPPO exhibits the poorest performance, failing to effectively handle agent heterogeneity and asymmetric roles. These results indicate that TAM-HAPPO maintains superior learning performance under the same adversarial conditions, suggesting that the observed improvement primarily stems from the proposed cooperative learning framework rather than differences in opponent strategies. Furthermore, the results demonstrate that TAM-HAPPO remains effective in scenarios with increased agent numbers and coordination complexity, highlighting its suitability for heterogeneous manned–unmanned cooperative air combat environments.

Fig. 14 illustrates the air combat trajectories of the Red and Blue forces in a 5v4 environment. At the start of the engagement, the Red MAV was positioned in the center of the combat formation, slightly higher than the four UAVs, providing information support to the UAVs while ensuring its own safety. The Blue's four UAVs formed an inverted trapezoidal formation, flying directly opposite the Red forces. Each UAV carried two missiles as offensive weapons. At  $t = 22$  s, both sides' forward-deployed UAVs formations detected and locked onto the enemy aircraft, simultaneously maneuvering to evade incoming missiles. The Red MAV retreated to the rear of the battlefield, prioritizing its own safety. At  $t = 55$  s, the first wave of fire ended. The Red forces, at the cost of one UAV, shot down the Blue's forward-deployed formation, and a second wave of engagement began. The remaining two Blue UAVs formations continued to pursue high-value Red MAV targets. The Red's three UAVs decisively turned around and encircled the Blue UAVs from the left, center, and right. At this point, the Blue UAVs attempted to turn and evade the engagement, but it was too late. Finally, at  $t = 79$  s, they were completely wiped out by the Red Team, and the battle ended.



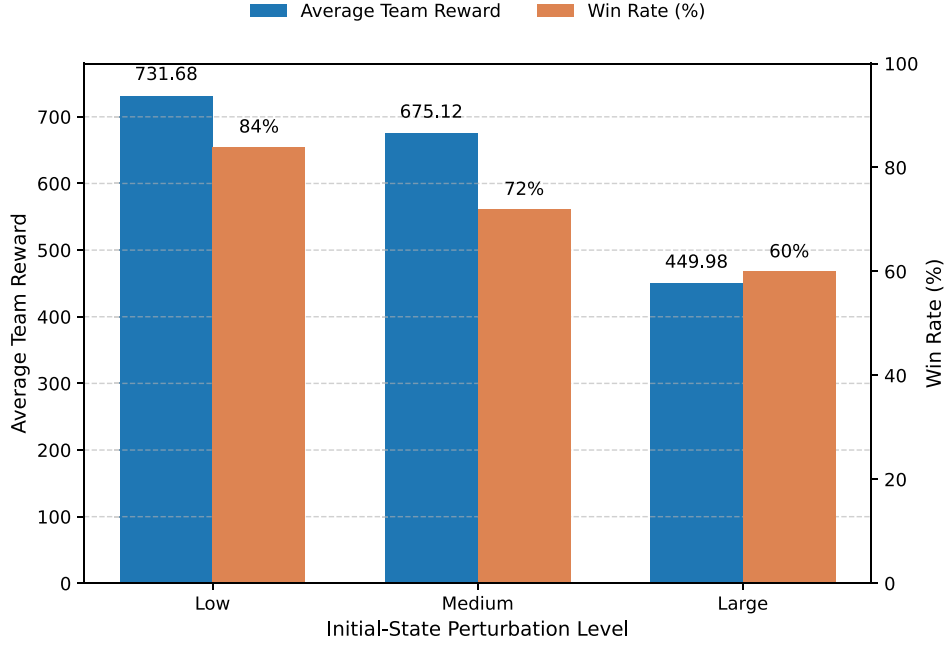


Fig. 16. Performance of TAM-HAPPO under different initial-state perturbation levels in the 5v4 environment.

#### 4.4. Ablation experiments

To evaluate the contribution of each proposed module to the overall performance of the algorithm, an ablation study was conducted. Specifically, we examined the impact of removing the state memory module (State Memory), the mask mechanism (Mask), and the attention mechanism (Attention), respectively. As shown in Fig. 15, the TAM-HAPPO curve (in blue) represents the full model incorporating all enhancements, while the other curves correspond to the variants with individual modules ablated. The baseline HAPPO algorithm is shown in orange.

As illustrated in the figure, the full TAM-HAPPO model achieves superior performance throughout the training process, exhibiting a rapid increase in episode reward during the early stages and maintaining a consistently high level as training progresses. This demonstrates the effectiveness of the proposed modules in enhancing policy learning and task execution in multi-agent air combat environments.

The contribution of each module is analyzed as follows:

- **State Memory Module (No\_State\_Memory, red curve):** Removing the state memory module results in significant instability during training and a noticeable drop in final performance. This highlights the importance of temporal modeling in the decision-making process. The State Memory module, based on a gated recurrent unit (GRU) structure, encodes sequential observations into compact representations capturing temporal dynamics. Additionally, it improves temporal consistency in the training dataset by ensuring orderly storage of experiences in the replay buffer, which in turn enhances training stability and sample efficiency.
- **Mask Mechanism (No\_Mask, purple curve):** The variant without the mask mechanism exhibits higher volatility and slower convergence. Without the ability to dynamically suppress the outputs of terminated agents, the network is more susceptible to disruptive influences from non-functional entities. The mask mechanism plays a critical role by filtering out invalid action outputs, ensuring that only active agents contribute to the cooperative policy learning, thus improving both the robustness and convergence speed of the algorithm.
- **Attention Mechanism (No\_Attention, brown curve):** Removing the attention module results in moderately degraded performance compared to the full model. The multi-head attention mechanism

incorporated in the value network enables the agent to better model inter-agent dependencies, enhancing value estimation accuracy in complex multi-agent interactions. Its absence reduces the representational capacity of the value function, leading to suboptimal performance.

- **Baseline HAPPO (orange curve):** Serving as the reference method, the baseline HAPPO algorithm shows the lowest reward accumulation and slowest learning rate among all variants. The lack of temporal modeling, dynamic masking, and attention mechanisms limits its ability to handle the high-dimensional, dynamic interactions inherent in multi-agent air combat environments.

In summary, each of the proposed modules contributes meaningfully to the algorithm's performance, with the state memory and mask mechanisms being particularly impactful. The ablation results substantiate the effectiveness of the TAM-HAPPO algorithm in enhancing learning stability, coordination capability, and overall performance in complex tactical environments.

#### 4.5. Generalization experiment

To evaluate the generalization capability and robustness of the proposed TAM-HAPPO algorithm under uncertain deployment conditions, a generalization experiment is conducted by introducing random perturbations to the initial states of all agents. Rather than modifying the policy structure or observation models, uncertainty is injected directly into the initial conditions at the beginning of each episode, which reflects variations in aircraft and missile deployment as well as mission initialization.

Specifically, bounded random perturbations are applied to the initial position, velocity, altitude, and heading of MAVs and UAVs. For each state variable, an independent uniformly distributed disturbance is added to its nominal initial value, which can be formulated as

$$\tilde{s}_0 = s_0 + \mathcal{U}(-\Delta, \Delta), \quad (30)$$

where  $s_0$  denotes the nominal initial state,  $\tilde{s}_0$  represents the perturbed initial state, and  $\Delta$  controls the magnitude of the initial uncertainty. This bounded perturbation scheme captures practical deployment errors while avoiding extreme outliers that may dominate the evaluation

results. To further investigate robustness under different levels of uncertainty, the 5v4 heterogeneous air combat scenario is selected as a representative test case, and three perturbation levels are designed, corresponding to low, medium, and large disturbances. The detailed perturbation ranges for each level are summarized in Table 8.

Without any additional fine-tuning, the trained policies are directly evaluated under randomized initial conditions. For each perturbation level, the heterogeneous air combat scenario is tested over 50 independent runs. Key performance metrics, including win rate and cumulative team reward, are collected and statistically analyzed to assess the stability and consistency of the learned policies.

The experimental results are illustrated in Fig. 16. Under the low perturbation setting, the proposed method achieves a win rate of 84% with an average cumulative team reward of 731.68, demonstrating good generalization to mildly perturbed initial conditions. As the perturbation level increases to medium and large settings, the win rate decreases to 72% and 60%, while the average cumulative reward drops to 675.12 and 449.98, respectively, reflecting the increased task difficulty caused by larger deployment uncertainties. Despite the performance degradation under severe perturbations, the proposed TAM-HAPPO algorithm preserves effective cooperative behaviors among MAVs, UAVs, and missiles, without catastrophic failure or loss of coordination. Overall, these results demonstrate that the proposed TAM-HAPPO framework exhibits strong robustness and generalization capability under varying degrees of initial-state uncertainty. Even when the initial deployment deviates significantly from the training configuration, the learned heterogeneous cooperation strategies remain stable and effective, highlighting the practical potential of the proposed method in complex air combat simulation environments.

## 5. Conclusions

This paper investigates cooperative air combat maneuver decision-making in heterogeneous environments involving MAVs, UAVs, and missiles, and proposes a TAM-HAPPO-based learning framework under the centralized training and decentralized execution paradigm. By integrating a state memory module, agent masking mechanism, entropy regularization, and a multi-head attention structure, the proposed method effectively alleviates policy coupling and partial observability issues among heterogeneous agents, resulting in improved training stability and cooperative performance. Extensive experiments in both homogeneous and heterogeneous combat environments demonstrate that the proposed approach ensures MAV survivability while enabling effective coordination among UAVs and missiles, and achieves higher cumulative team rewards and superior overall combat performance compared with representative baseline methods.

Future work will be pursued along two main directions. First, practical deployment factors such as communication constraints, sensing uncertainty, and control delays will be explicitly considered to further evaluate the robustness of the proposed framework under more complex and practical simulation conditions. Second, learning-based adversaries and self-play training mechanisms will be introduced to construct more challenging combat environments and further enhance policy optimality and generalization capability.

## CRedit authorship contribution statement

**Caiyi Chen:** Writing – original draft, Writing – review & editing, Methodology; **Delin Luo:** Methodology, Supervision; **Zhengyu Guo:** Methodology, Funding acquisition.

## Data availability

Data will be made available on request.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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